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<https://russelldudek.github.io/ByteCubit/>

Legacy and target paths run side by side before cutover.

LEGACY FUNCTION RECORD

Purpose

Which production decision, action, record, or compliance obligation does the function support?

Users

Operators, quality, maintenance, engineering, support, R&D, analytics, customers?

Inputs

Tags, sensors, manual inputs, schedules, recipes, SQL, external systems?

Logic

States, calculations, queries, scripts, transactions, and exception handling?

Consumers

Displays, reports, SQL jobs, PI points/AF, alerts, data lake, or external applications?

Consequence

What happens when the function is wrong, late, unavailable, or ambiguous?

Owner

Who accepts behavior, data, release, and ongoing support?

TARGET TWIN

Ignition application

Project/component boundaries, reuse, navigation, state, diagnostics, and security.

Tag/UDT model

Naming, types, metadata, alarms, quality, ownership, and versioning.

SQL interaction

Named queries, transactions, schema, error handling, performance, and access.

Historian context

Point mapping, asset hierarchy, units, timestamps, quality, retention, and consumers.

Cloud contract

Schema, semantics, freshness, quality, access, owner, and intended consumers.

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Release Decision | Promote, Hold, Repair, or Retire

ACCEPTANCE MATRIX

Function parity

Normal states, transitions, calculations, commands, and exceptions.

Data integrity

Names, values, units, timestamps, quality, completeness, and reconciliation.

Operator usability

Task flow, visibility, alarms, acknowledgment, recovery, and training.

Performance

Response, query/load behavior, store-forward, and degradation handling.

Security and access

Roles, credentials, least privilege, network path, and auditability.

Supportability

Diagnostics, logs, runbook, owner, escalation, source, and version.

Rollback

Trigger, authority, procedure, data reconciliation, and communication.

Observation

Window, measures, owner, incident path, and learning capture.

DECISION RECORD

Promote

Evidence supports cutover; ownership and rollback are explicit.

Hold

Evidence is incomplete or production timing/conditions are wrong.

Repair

Target behavior, data context, usability, performance, or support needs change.

Retire

The legacy function is not worth reproducing; remove it with consumer approval.

LEARNING RETURN

What component, naming rule, test, diagnostic, runbook, data contract, or governance decision can the next modernization tranche reuse?